

[HOME](#) [TOP](#) [CATALOG](#) [CONTESTS](#) [GYM](#) [PROBLEMSET](#) [GROUPS](#) [RATING](#) [EDU](#) [API](#) [CALENDAR](#) [HELP](#)
[PROBLEMS](#) [SUBMIT CODE](#) [MY SUBMISSIONS](#) [STATUS](#) [STANDINGS](#) [CUSTOM INVOCATION](#)

G. 220 Trees

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

There is a Binary Tree with N nodes. You are given the in-order and pre-order traversals of the tree. Your task is to determine the post-order traversal of the tree.

Input

The first line contains an integer N ($1 \leq N \leq 1000$) — the number of nodes in the binary tree.

In the next line, there will be N integers $a_1, a_2, a_3 \dots a_n$ ($1 \leq a_i \leq N$) separated by spaces — representing the in-order traversal of the tree.

The following line, there will be N integers $b_1, b_2, b_3 \dots b_n$ ($1 \leq b_i \leq N$) separated by spaces — representing the pre-order traversal of the tree.

Output

Print N space-separated integers representing the post-order traversal of the binary tree.

Example

input	Copy
5 4 2 5 1 3 1 2 4 5 3	
output	Copy
4 5 2 3 1	

CSE221 Assignment 03 Summer 2026

Contest is running

24:27:51

Contestant



→ Submit?

Language: Java 21 64bit

Choose file:

Choose File

No file chosen



Success!



Submit

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The only programming contests Web 2.0 platform

Server time: Jul/11/2026 13:32:06^{UTC+6} (g1).

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